



Employee Attrition Prediction

By Group - **M4**

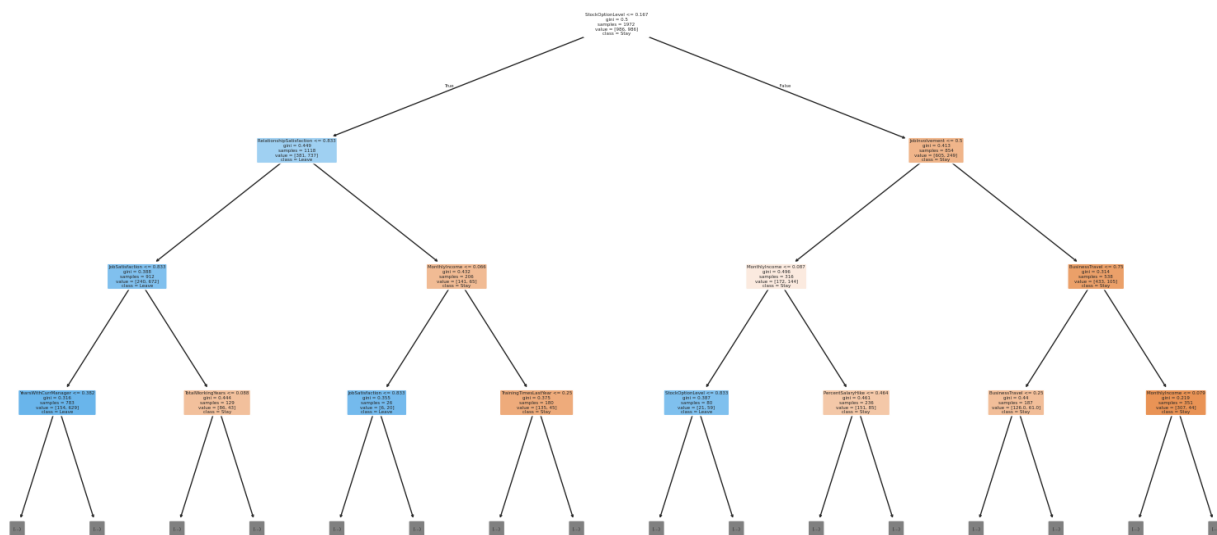
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Sample Data

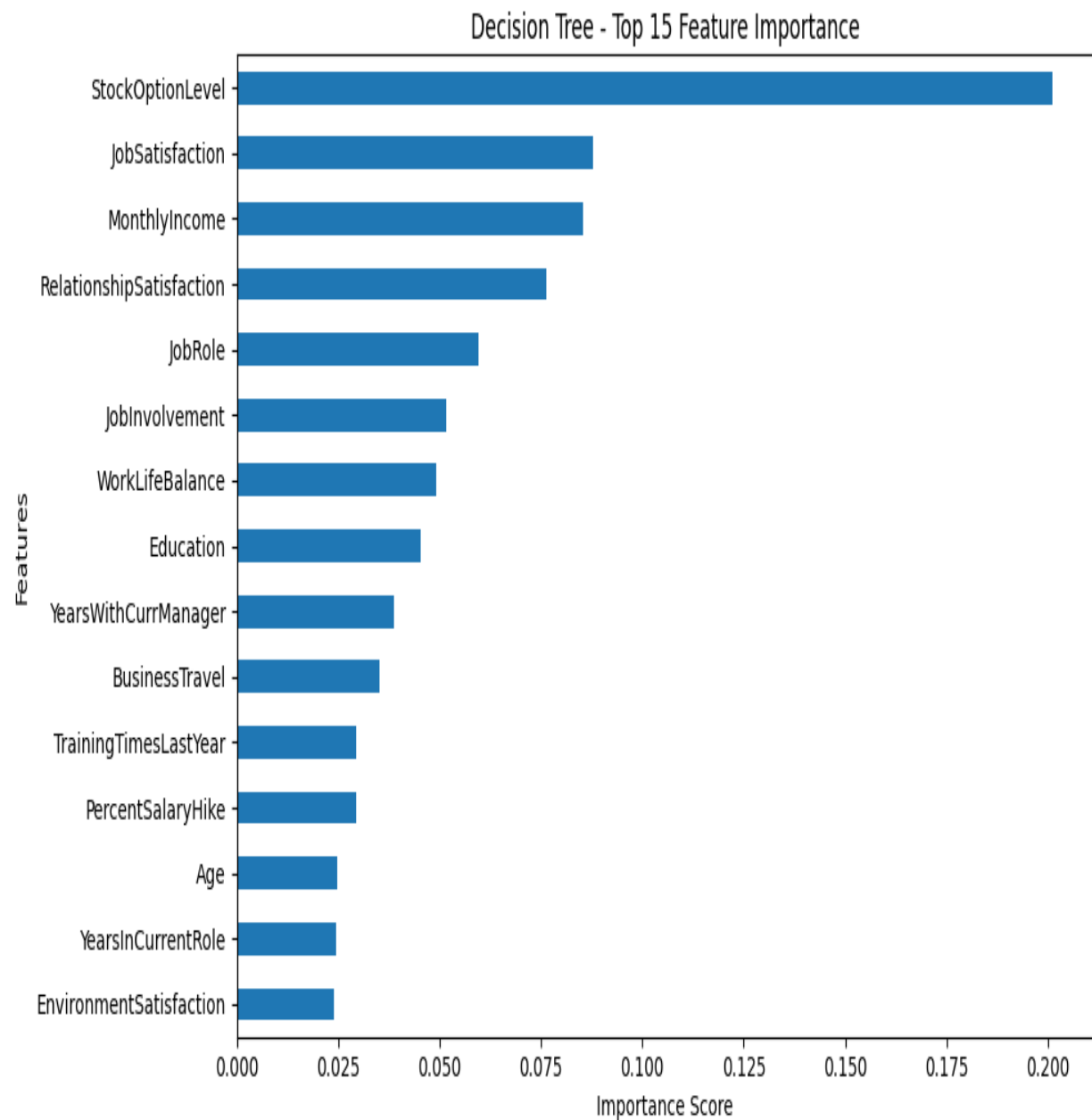
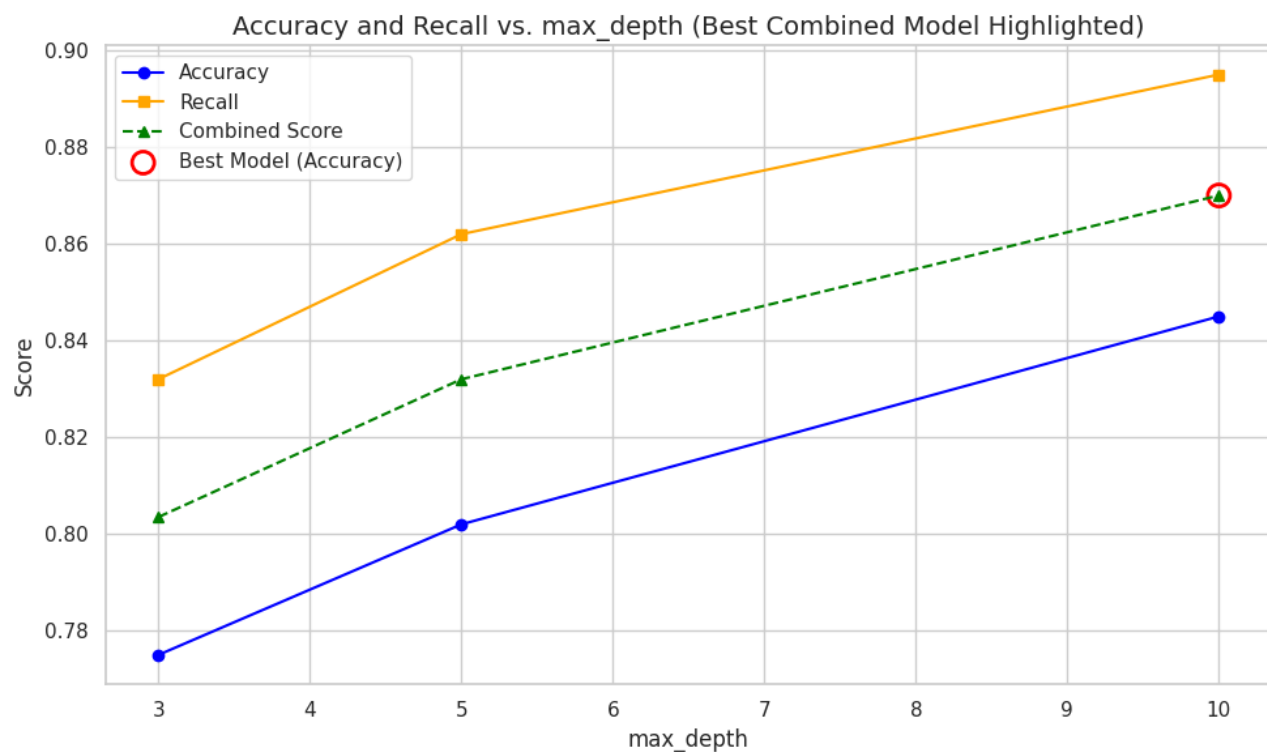
Age	Attrition	Business_Travel	Daily_Rate	Department	Distance_From_Home	Education	Education_Field	Employee_Count	Employee_Number	Environment_Satisfaction	Gender	HourlyRate	JobInvolvement	JobLevel	JobRole	JobSatisfaction	MaritalStatus	MonthlyIncome
41	Yes	avel_Rare	1102	Sales	1	2	Life Science	1	1	2	Female	94	3	2	Sales Executive	4	Single	5993
49	No	rel_Freque	279	rch & Develo	8	1	Life Science	1	2	3	Male	61	2	2	Research Scientist	2	Married	5130
37	Yes	avel_Rare	1373	rch & Develo	2	2	Other	1	4	4	Male	92	2	1	aboratory Technician	3	Single	2090
33	No	rel_Freque	1392	rch & Develo	3	4	Life Science	1	5	4	Female	56	3	1	Research Scientist	3	Married	2909
27	No	avel_Rare	591	rch & Develo	2	1	Medical	1	7	1	Male	40	3	1	aboratory Technician	2	Married	3468
32	No	rel_Freque	1005	rch & Develo	2	2	Life Science	1	8	4	Male	79	3	1	aboratory Technician	4	Single	3068
59	No	avel_Rare	1324	rch & Develo	3	3	Medical	1	10	3	Female	81	4	1	aboratory Technician	1	Married	2670
30	No	avel_Rare	1358	rch & Develo	24	1	Life Science	1	11	4	Male	67	3	1	aboratory Technician	3	Divorced	2693
38	No	rel_Freque	216	rch & Develo	23	3	Life Science	1	12	4	Male	44	2	3	lanufacturing Directo	3	Single	9526
36	No	avel_Rare	1299	rch & Develo	27	3	Medical	1	13	3	Male	94	3	2	althcare Representat	3	Married	5237
35	No	avel_Rare	809	rch & Develo	16	3	Medical	1	14	1	Male	84	4	1	aboratory Technician	2	Married	2426
29	No	avel_Rare	153	rch & Develo	15	2	Life Science	1	15	4	Female	49	2	2	aboratory Technician	3	Single	4193
31	No	avel_Rare	670	rch & Develo	26	1	Life Science	1	16	1	Male	31	3	1	Research Scientist	3	Divorced	2911
34	No	avel_Rare	1346	rch & Develo	19	2	Medical	1	18	2	Male	93	3	1	aboratory Technician	4	Divorced	2661
28	Yes	avel_Rare	103	rch & Develo	24	3	Life Science	1	19	3	Male	50	2	1	aboratory Technician	3	Single	2028
29	No	avel_Rare	1389	rch & Develo	21	4	Life Science	1	20	2	Female	51	4	3	lanufacturing Directo	1	Divorced	9980
32	No	avel_Rare	334	rch & Develo	5	2	Life Science	1	21	1	Male	80	4	1	Research Scientist	2	Divorced	3298
22	No	Non-Travel	1123	rch & Develo	16	2	Medical	1	22	4	Male	96	4	1	aboratory Technician	4	Divorced	2935
53	No	avel_Rare	1219	Sales	2	4	Life Science	1	23	1	Female	78	2	4	Manager	4	Married	15427
38	No	avel_Rare	371	rch & Develo	2	3	Life Science	1	24	4	Male	45	3	1	Research Scientist	4	Single	3944

Decision Tree

Decision Tree Visualization (Top 3 Levels)



- Tree depths we tried = 3, 5, 10
- Top features – Stock Option Level, Job Satisfaction, Monthly Income, Relationship satisfaction and Job role
- Found the optimal tree depth to be at 10 with AUC = 0.81 and Recall = 0.86



Logistic Regression:

Forward Feature Selection: Accuracy=0.7976, AUC=0.8683, Recall= 0.8016

Backward Feature Selection: Accuracy=0.8057, AUC=0.8822, Recall = 0.8016

- Found Backward feature selection model to be the best one

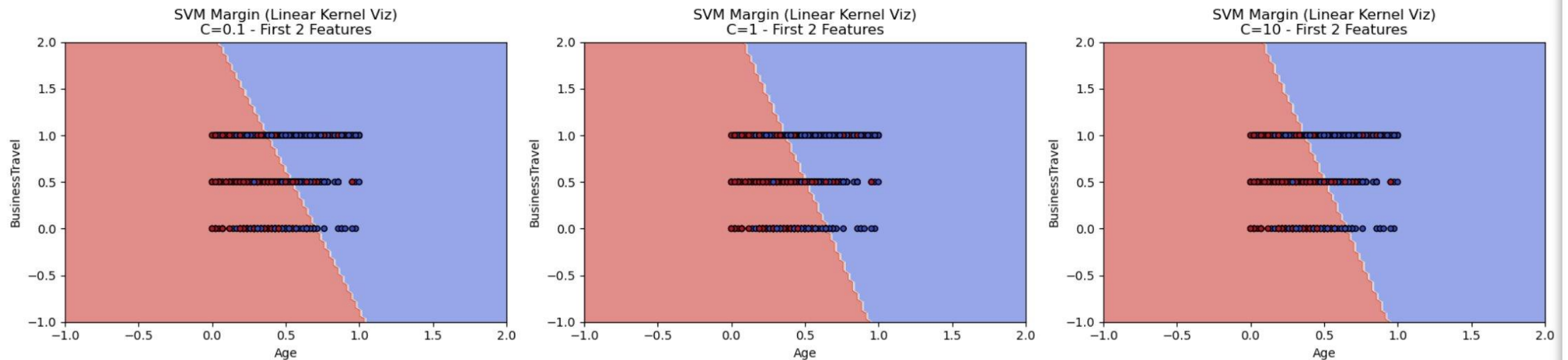
Insights from the top features:

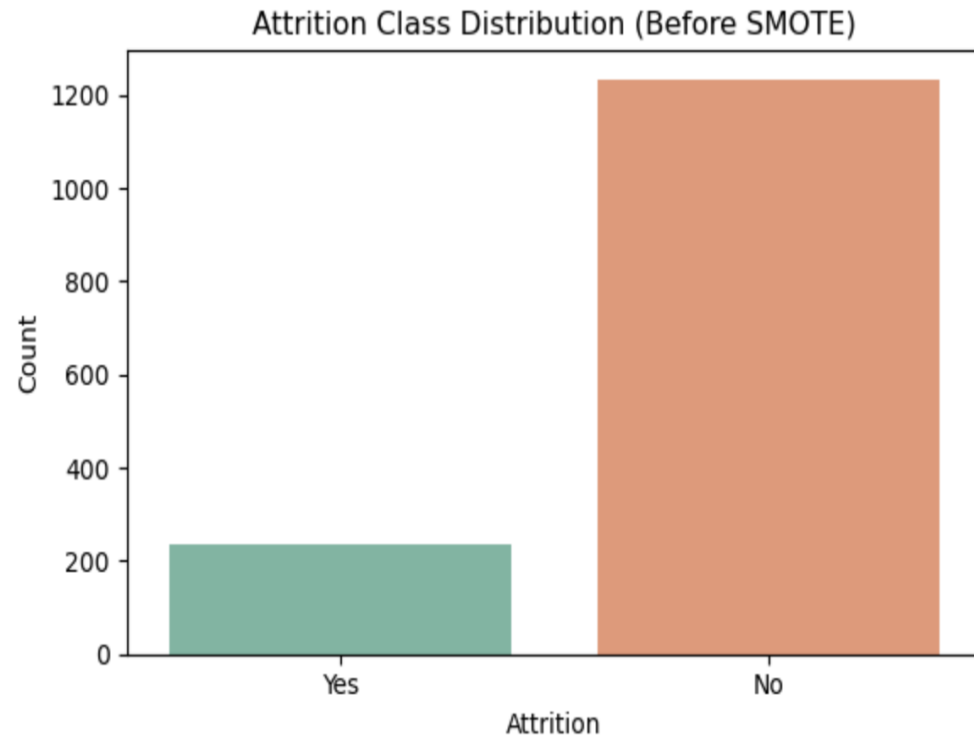
- Higher stock options make employees 2.5 times less likely to leave
- Employees who are more involved in their jobs are 2.2 times less likely to leave
- Higher job levels (more senior positions) reduce departure risk by about 2 times
- Working overtime increases the chance of leaving by 0.8 times
- Living farther from work increases departure risk by 1.2 times
- More years since last promotion significantly increases departure likelihood by 1.7 times

SVM(Support Vector Machine):

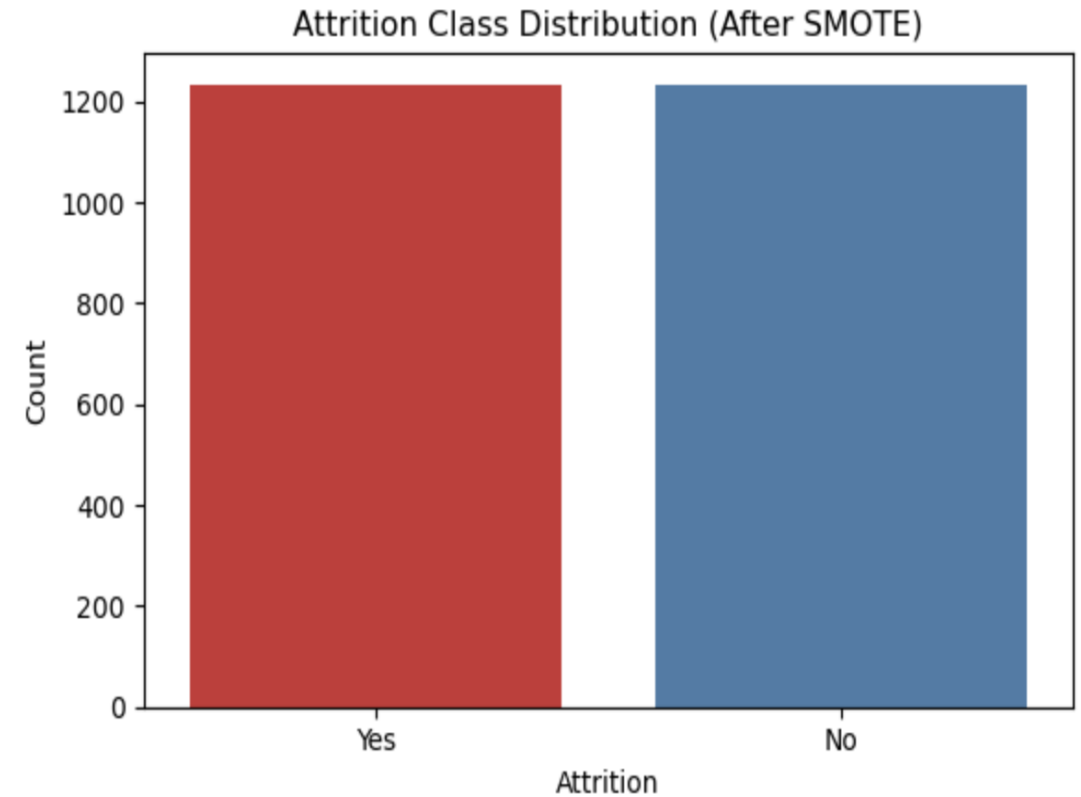
- Tried different **C** values: 0.1, 1, 10
- **Accuracy with different C values:** C=0.1: Accuracy=0.8410, C = 1: Accuracy = 0.8902, C = 10: Accuracy = 0.9191
- **Optimal C = 10** gave **AUC = 96.66, Accuracy = 91.91**

SVM Decision Boundary Visualization (Linear Kernel on First 2 Features)





Original class distribution:
Attrition
No 1233
Yes 237
Name: count, dtype: int64

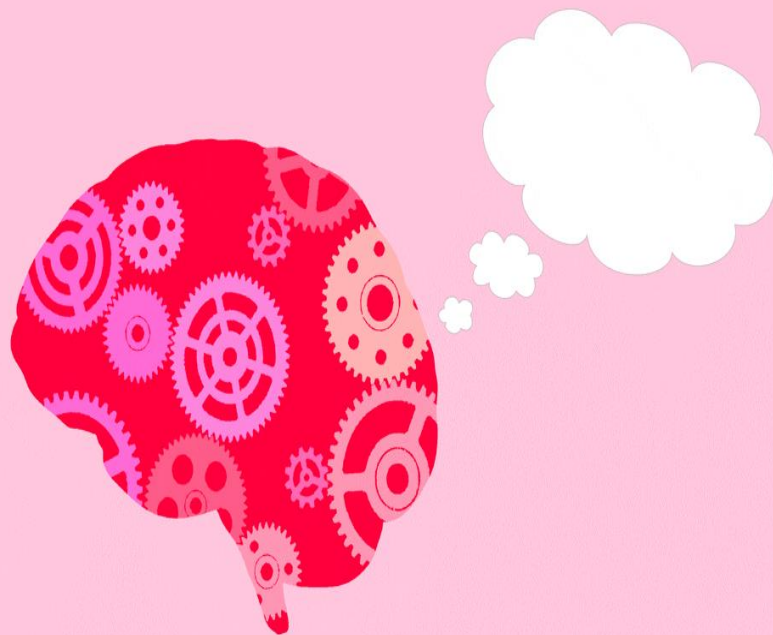


Class distribution after SMOTE:
Counter({1: 1233, 0: 1233})

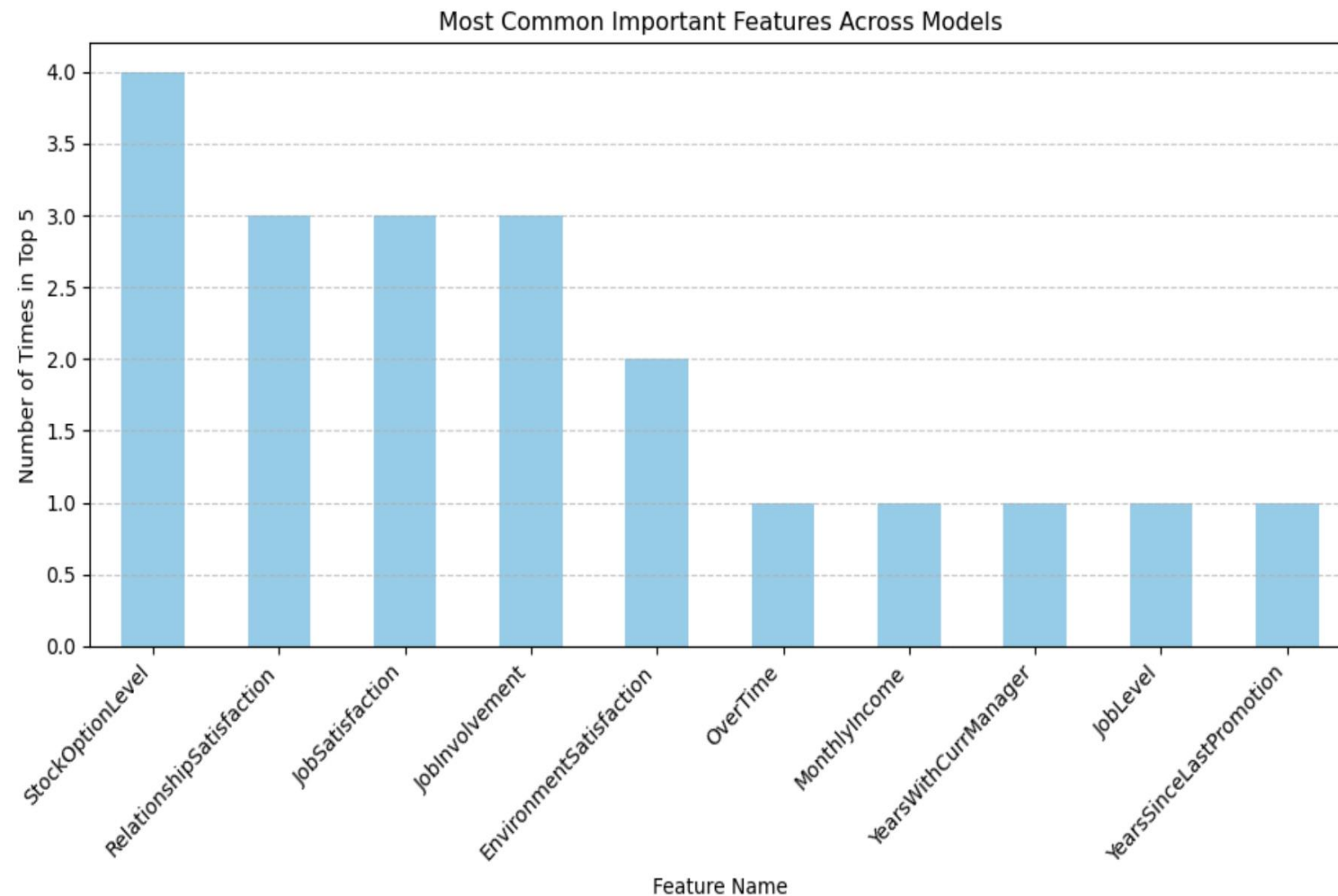
Dataset

Measure of Importance

Recall

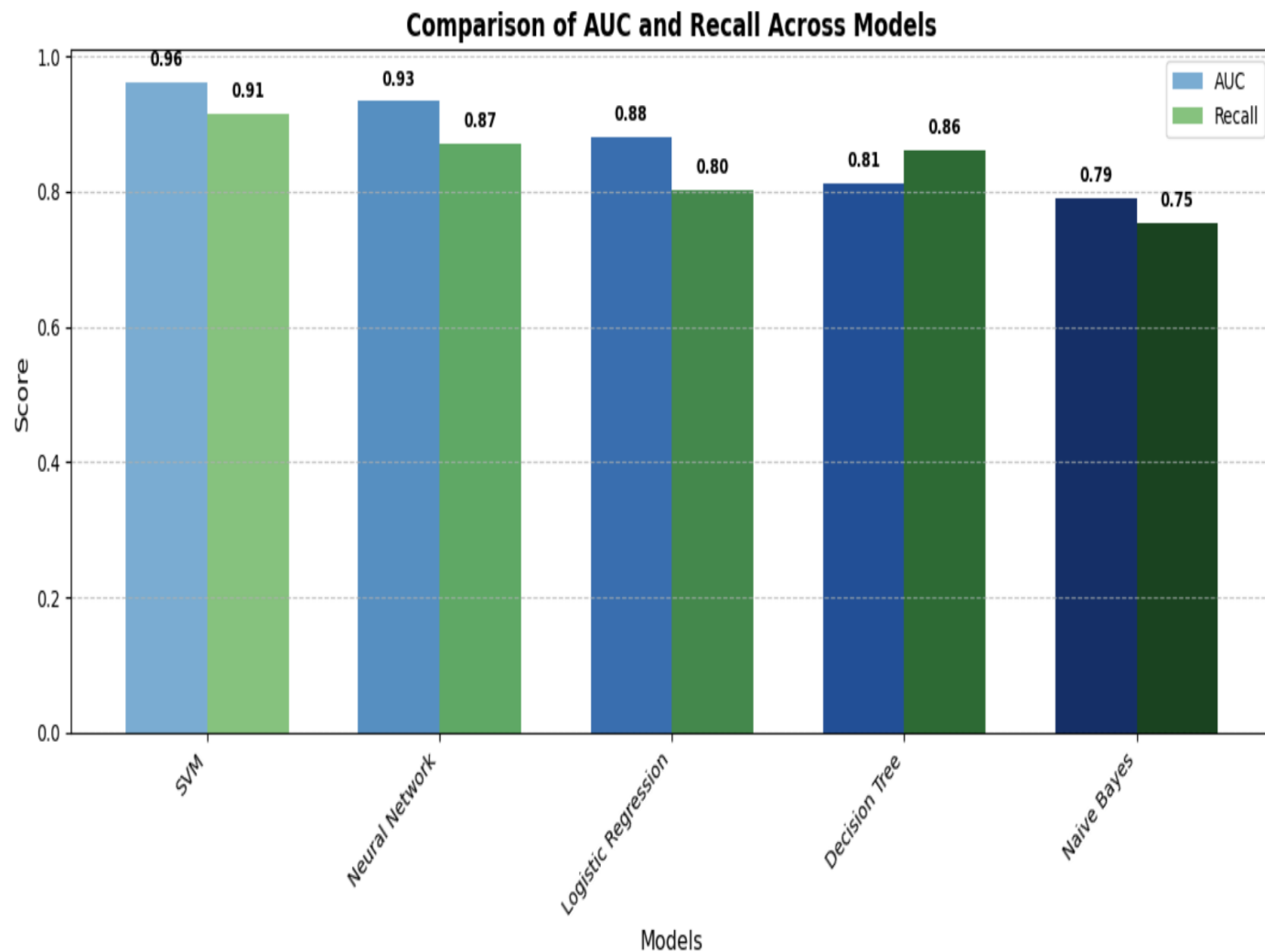


Top Features



So far..

Model	HyperParameters
SVM	C=10
Neural Network	hidden_layer_sizes': (100, 50), 'activation': 'relu'
Logistic Regression	Backward Elimination (17 features)
Decision Tree	max_depth=10
Naive Bayes	No Hyperparameters Tuned



Random Forest



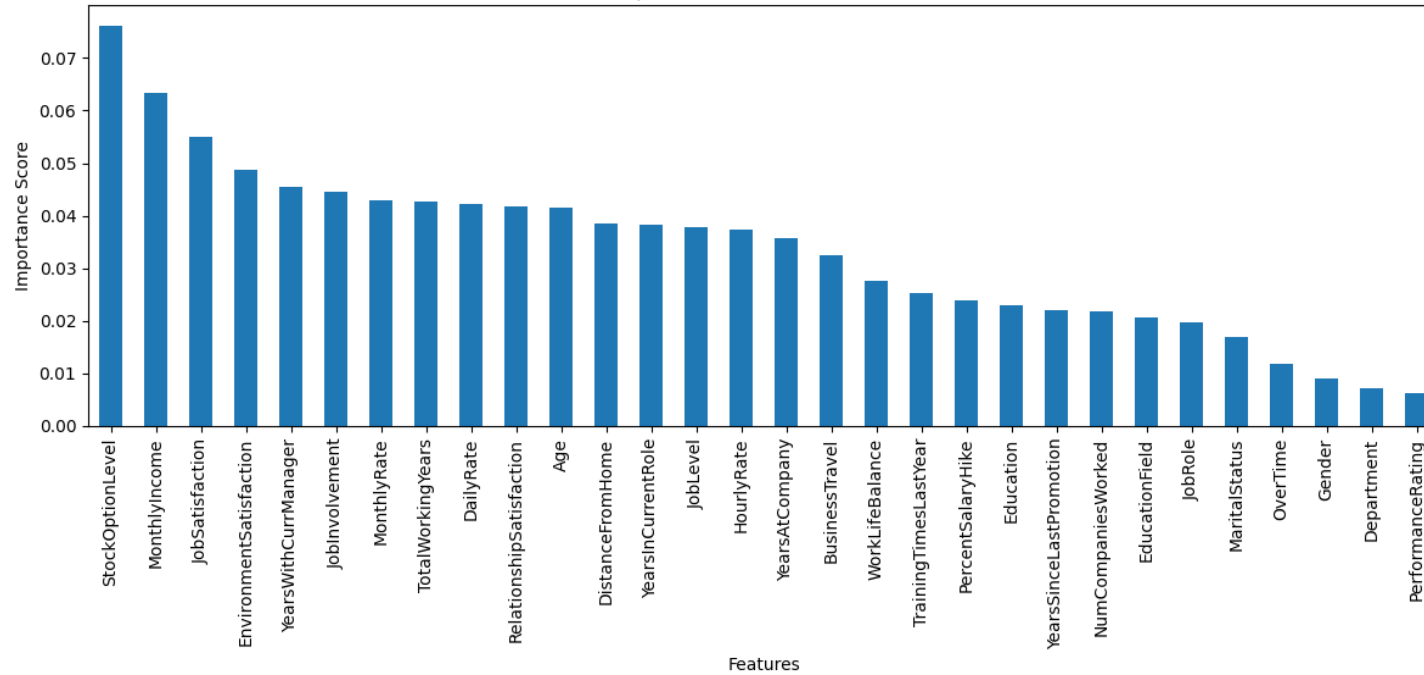
Performance Metrics:

- Accuracy: 88.9%
- AUC Score: 0.96
- Recall: 88.3% (Strong at identifying at-risk employees)

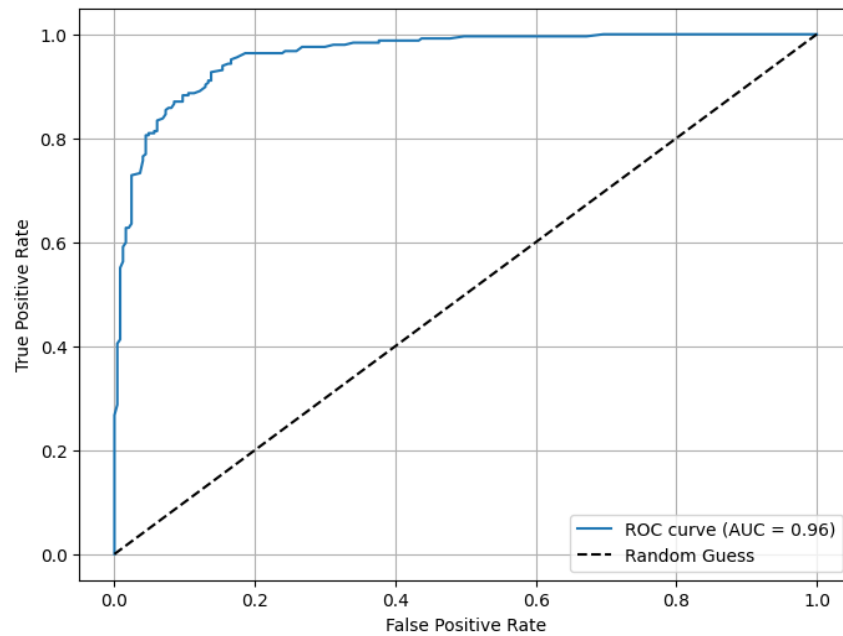
Optimal Parameters:

- 200 Estimators
- Max Depth = 20
- Min Samples Split = 2

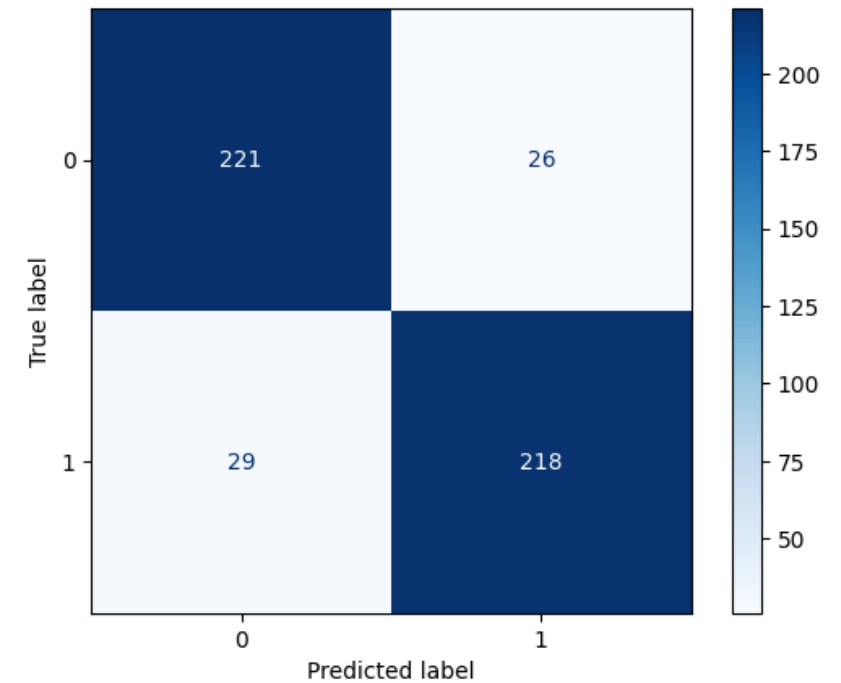
Feature Importances from Best Random Forest



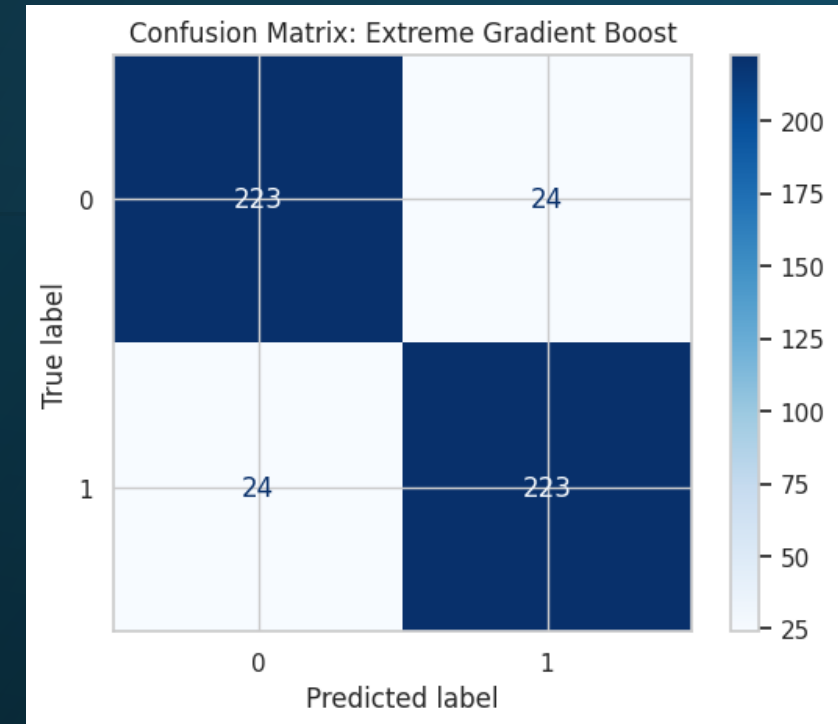
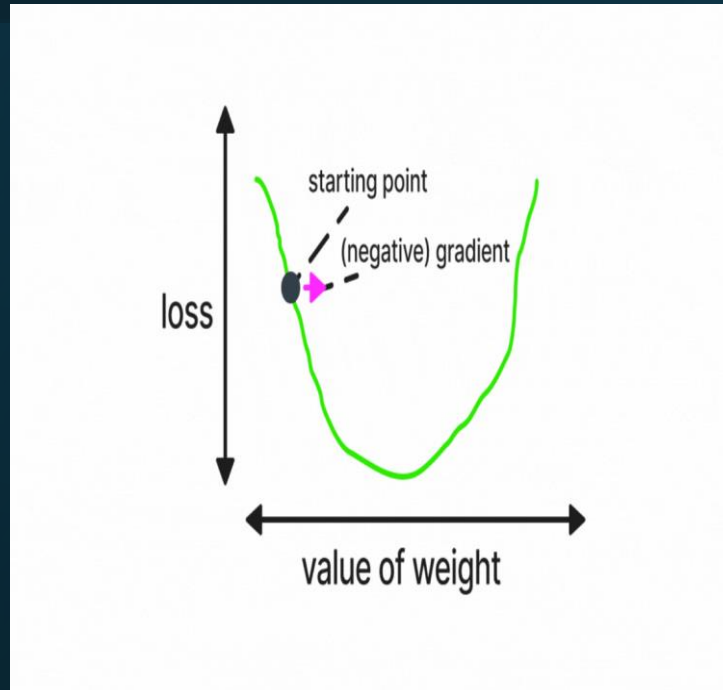
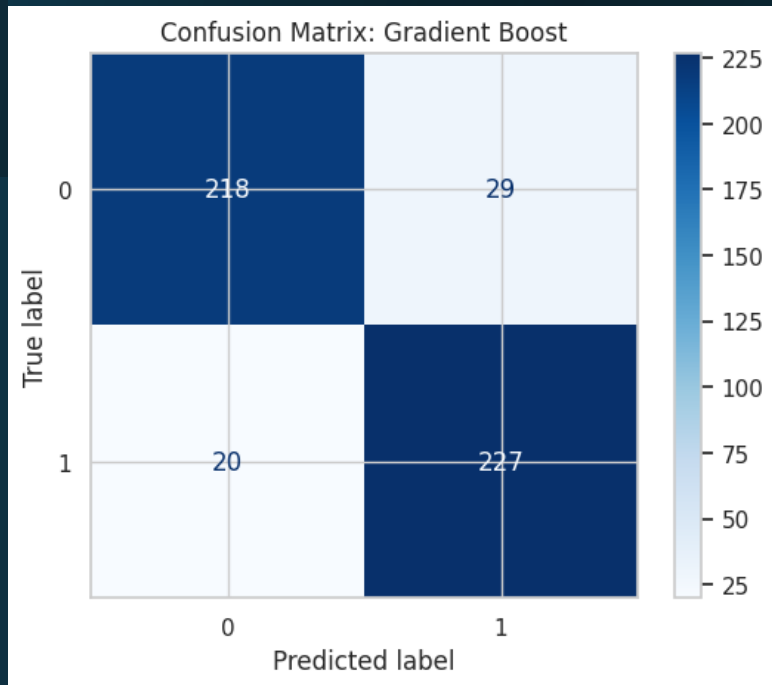
ROC Curve: Random Forest



Confusion Matrix: Random Forest



Gradient Boosting & XGBoost



Contrasting the two

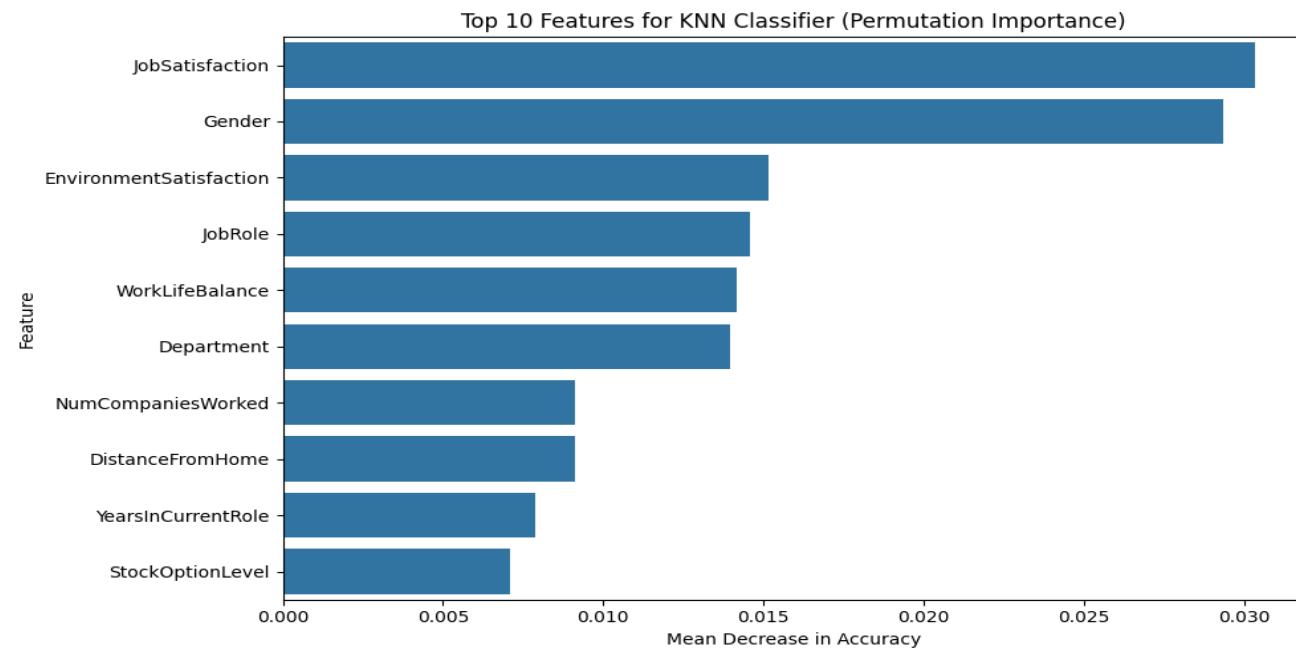
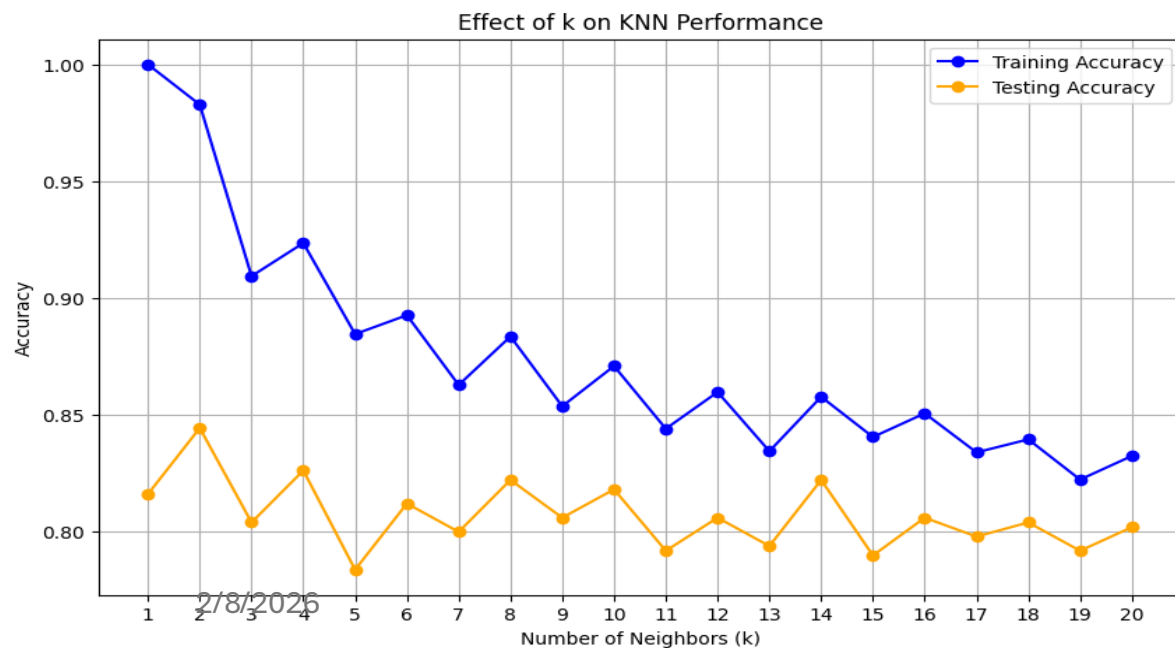
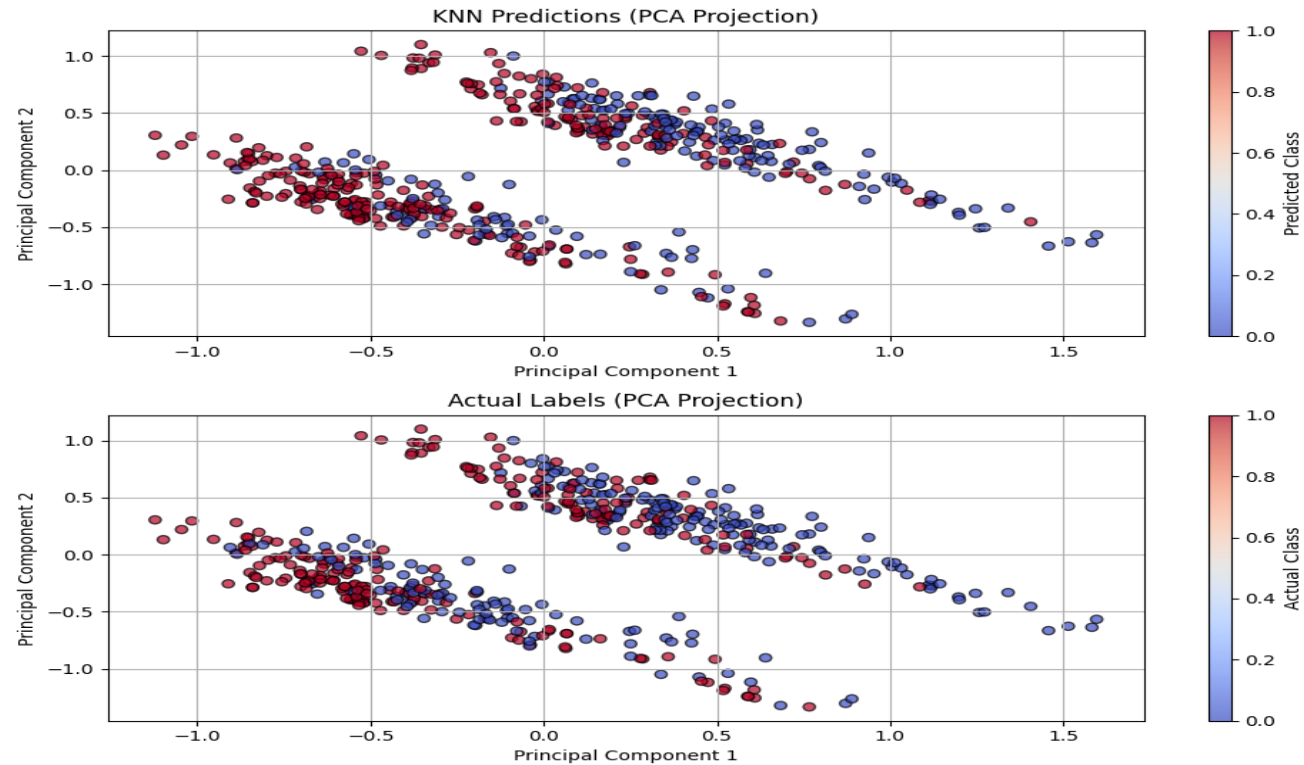
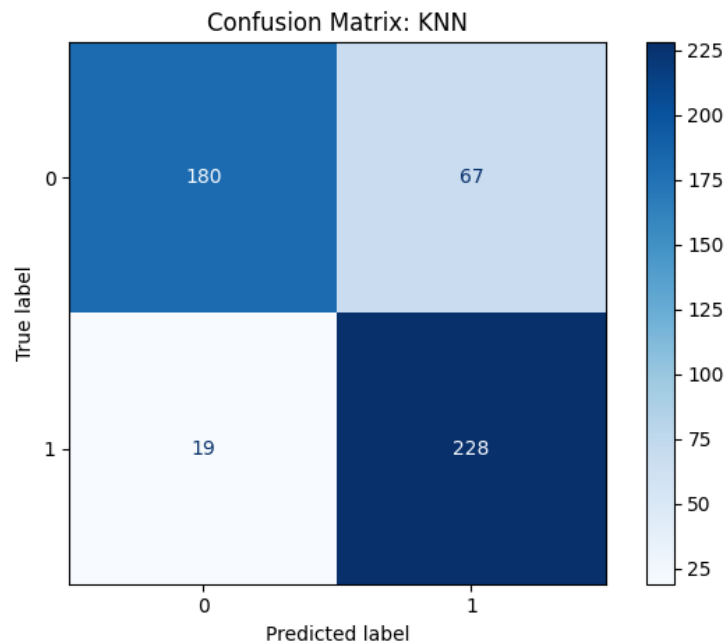
Aspect	Gradient Boost	XGBoost
Accuracy	90%	90%
Recall	92%	90%
Precision	88%	90%
AUC	97%	97.02%
Simplicity	Simpler Hyperparameters (3 tuned)	More Tuned Hyperparameters (6 tuned)
Regularization	None	L1 (alpha) and L2 (lambda) regularization
Feature Focus	Income & Satisfaction	Position, Career Progression

- **Gradient Boost** offers slightly better recall with a simpler setup.
- **XGBoost** provides stronger regularization, slightly better AUC, and more balanced generalization across features.

KNN

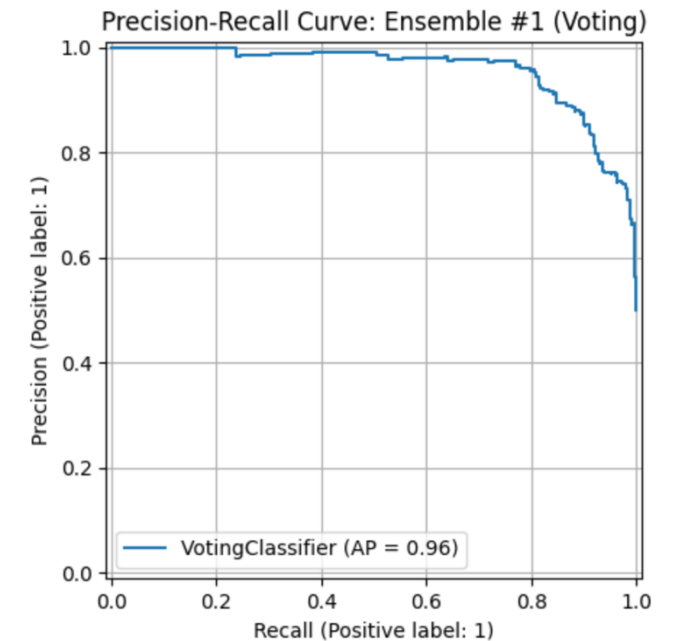
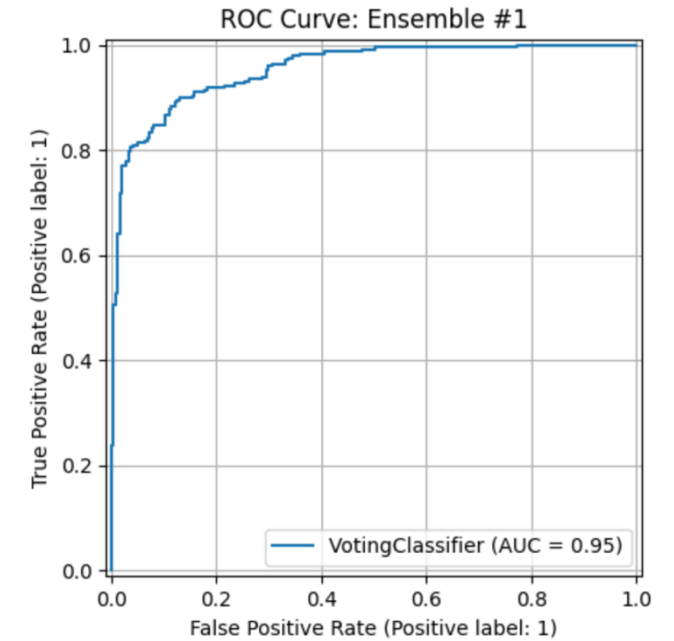
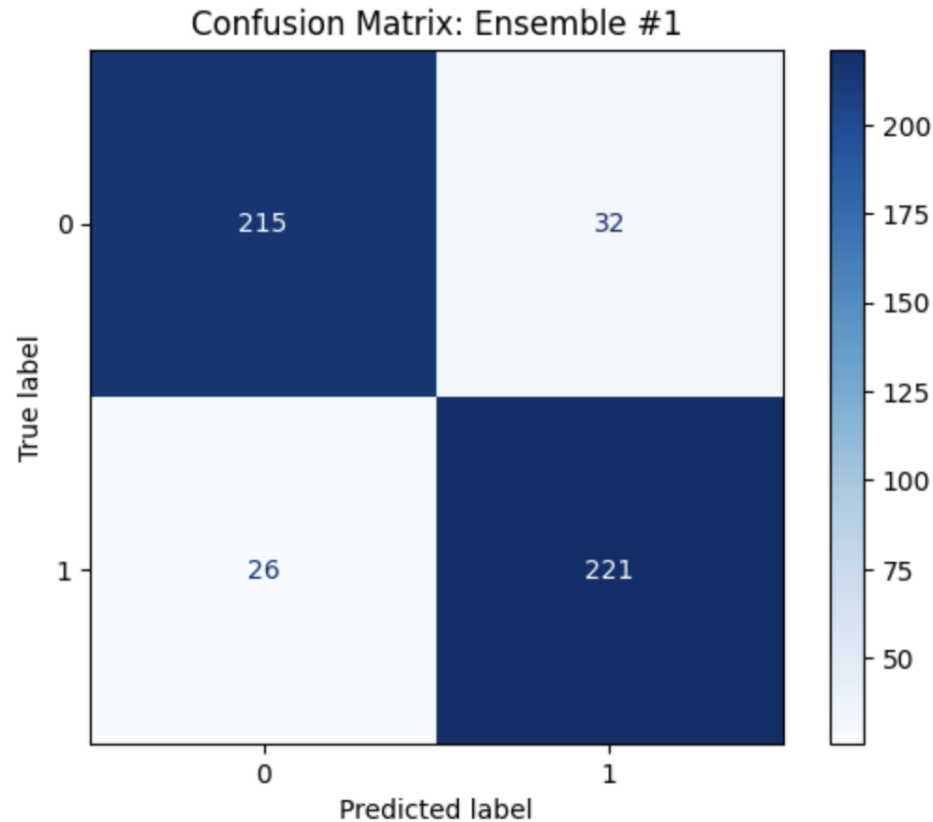
- KNN achieved 82.6% accuracy and 0.90 AUC score, making it one of our top performers
- Optimal model uses 4 nearest neighbors to make predictions
- Exceptional at identifying at-risk employees with 92.3% recall rate
- Feature importance analysis revealed JobSatisfaction and Gender as the key predictors
- Works well with our dataset without requiring complex parameter tuning
- Provides excellent balance between computational efficiency and predictive power

K-NN



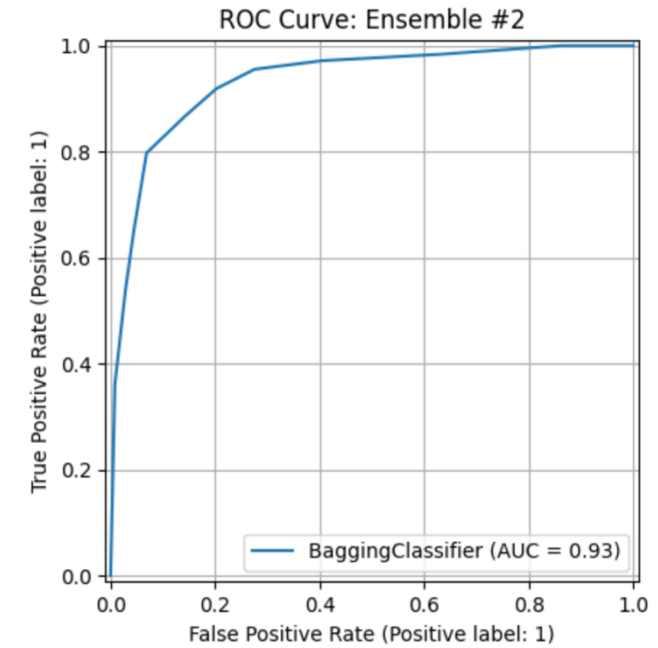
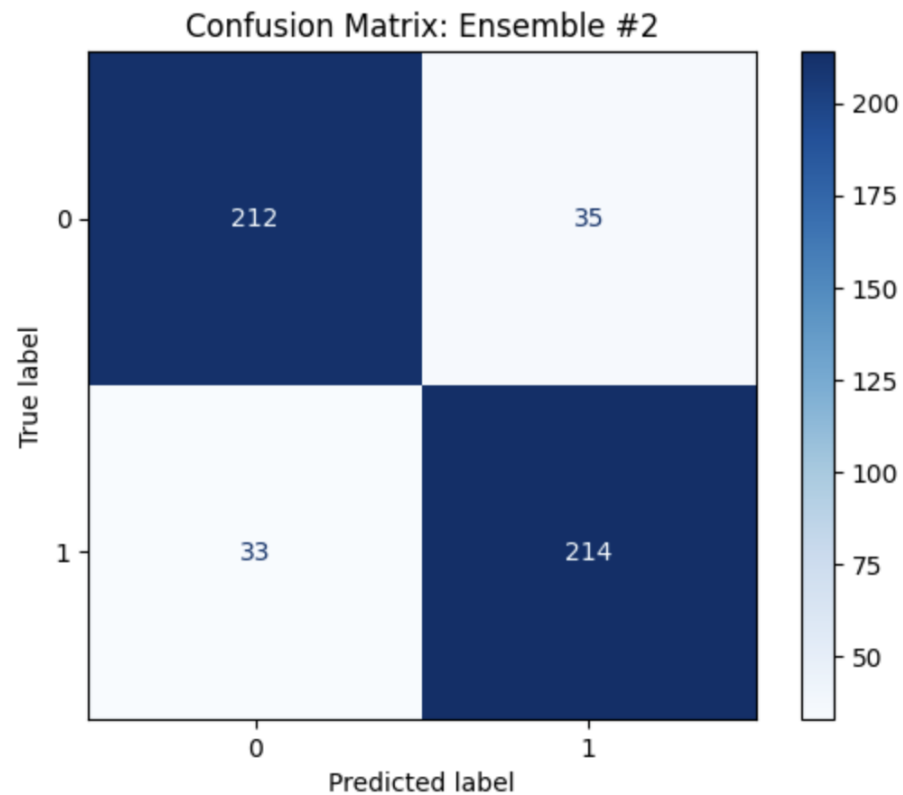
Ensemble Method #1 (SVM, DT, NN)

- 'Accuracy': 0.88
- 'Precision': 0.87
- 'Recall': 0.89
- 'AUC': 0.95



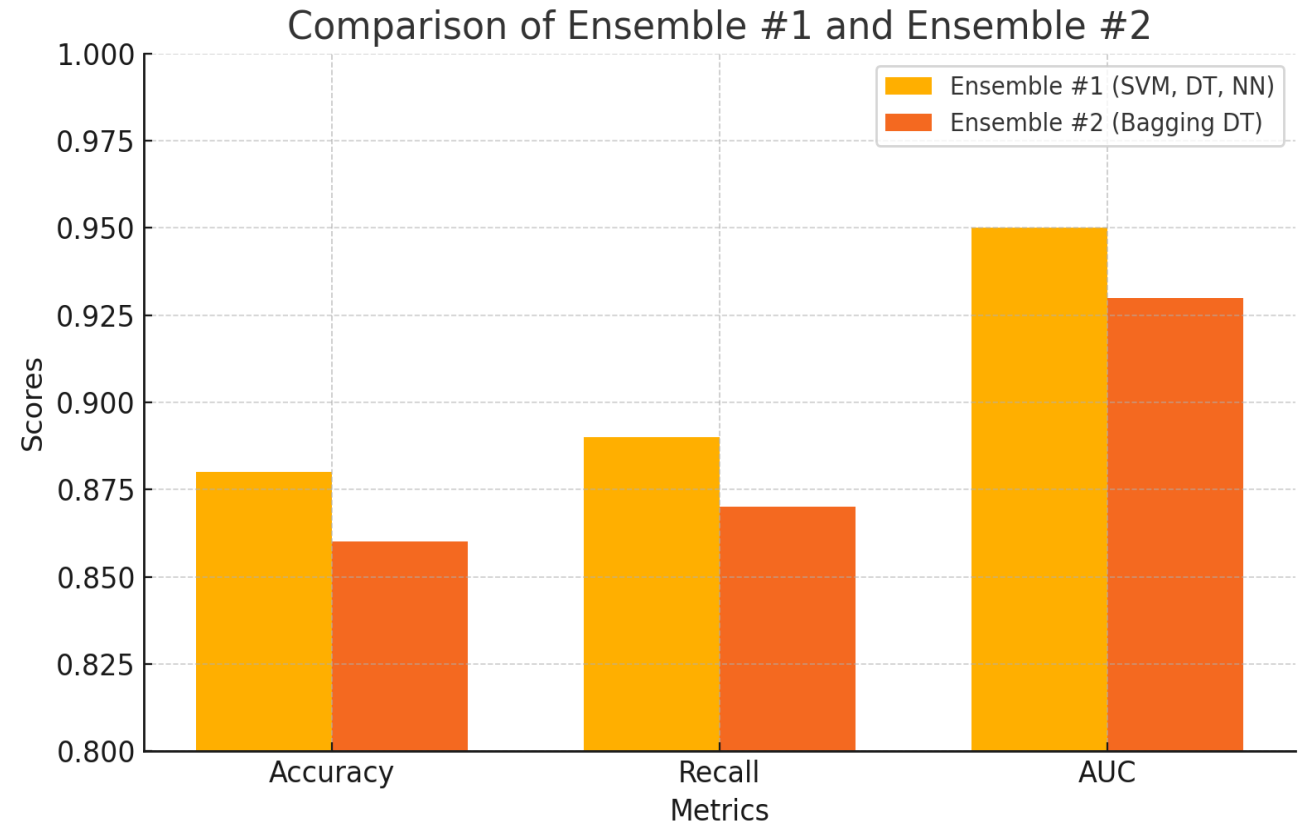
Ensemble Method #2 (Bagging with Decision Trees)

- 'Accuracy': 0.86
- 'Precision': 0.86
- 'Recall': 0.87
- 'AUC': 0.93



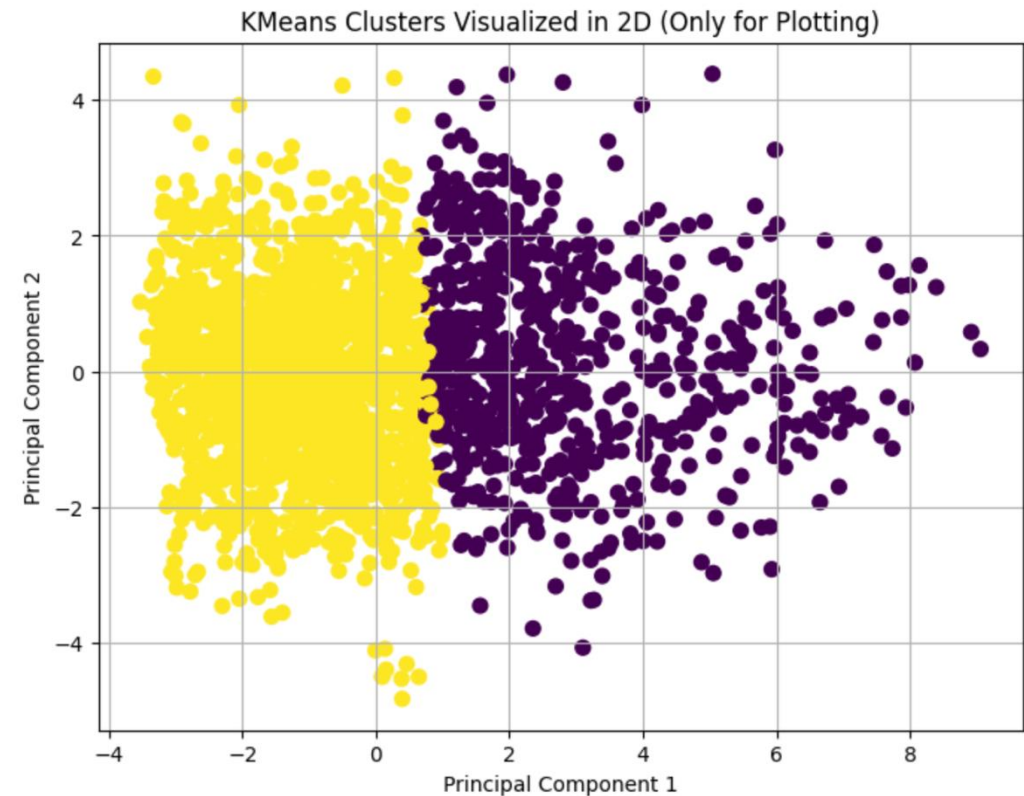
Comparison of 2 Ensemble models

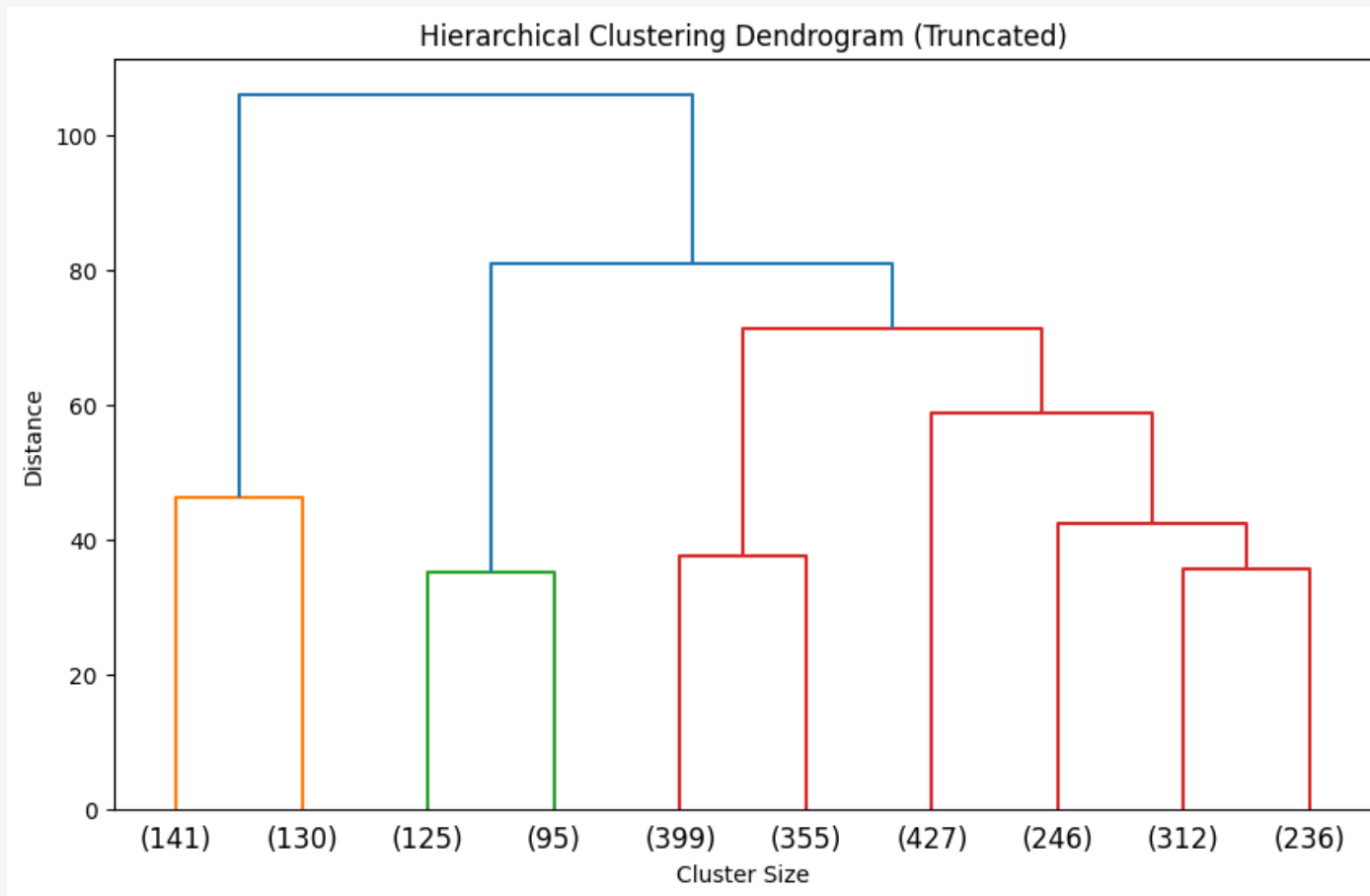
- Ensemble #1 has **2% higher accuracy**, meaning it correctly predicts attrition slightly better overall
- Ensemble #1 captures **more true positives** (employees leaving) — important for the business because missing leavers is costly.
- Ensemble #1 again shows **stronger separation** between employees who leave vs. those who stay.



K-Means Clustering Visualization (k=2)

- Clustering performed **directly on scaled features** without PCA.
- Optimal number of clusters (**k=2**) selected using **Silhouette Score** and **elbow method**.
- Achieved a **Silhouette Score of 0.1023**, indicating moderate cluster separation.
- Clusters show **reasonable but overlapping boundaries**, reflecting real-world data complexity.
- PCA was initially considered but **discarded** after performance comparison.





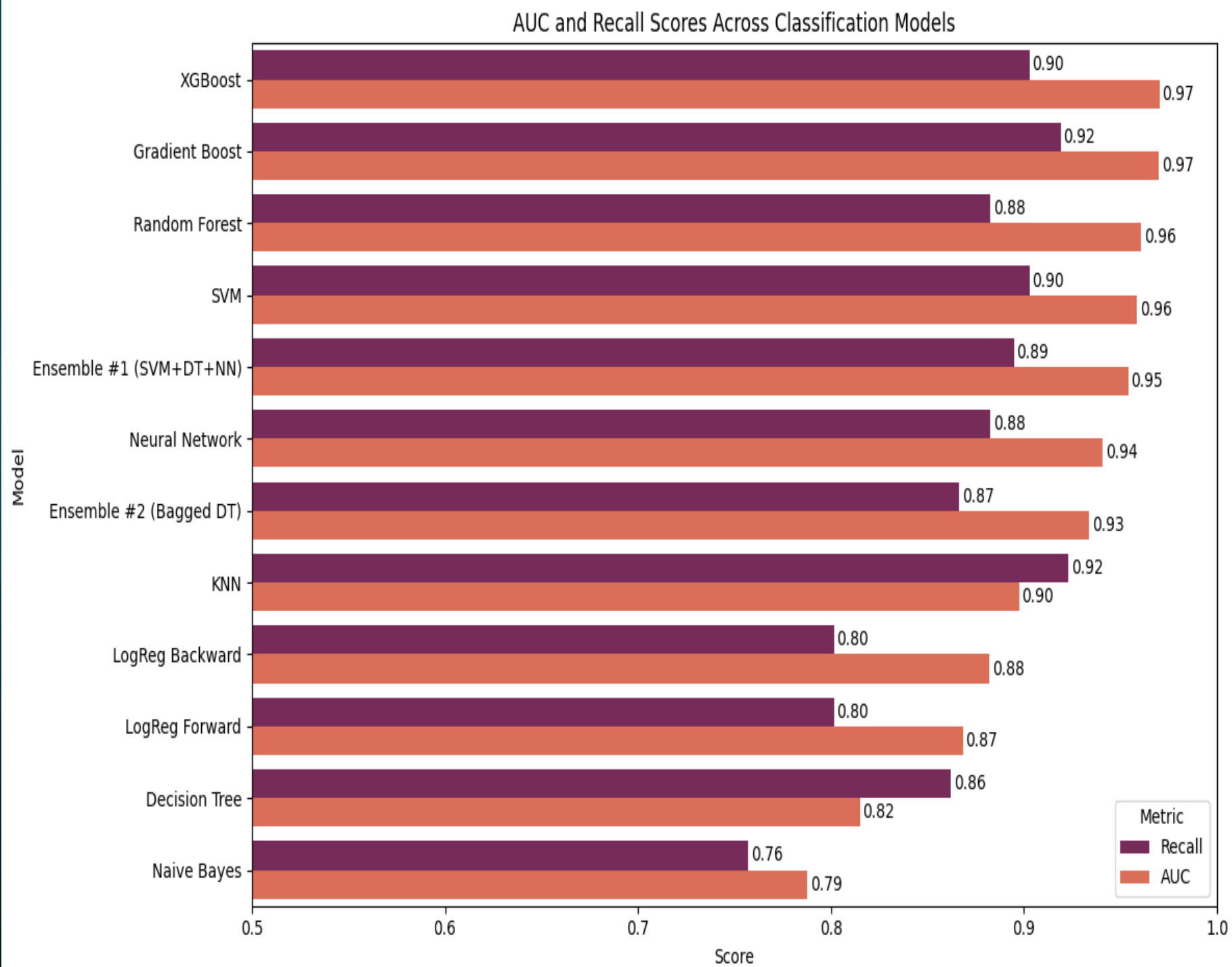
Agglomerative (Hierarchical) Clustering

- **Agglomerative Clustering** applied with `n_clusters = 2`.
- **Silhouette Score: 0.0726** — even lower than KMeans, meaning the cluster separation is weaker.
- **Visualization:**
 - **Dendrogram** built using Ward's method.
- Shows cluster merging patterns; large vertical gaps suggest natural cluster separation points.

Chosen Hyperparameters for each model

	Model	Best Hyperparameters
0	Random Forest	n_estimators=200, max_depth=20
1	Extreme Gradient Boost	n_estimators=200, max_depth=5, learning_rate=0.1
2	Gradient Boost	n_estimators=200, max_depth=5, learning_rate=0.1
3	Ensemble #1 (SVM+DT+NN)	Soft voting (SVM, DT, NN)
4	Neural Network	hidden_layers=(100,), activation=relu, solver=adam
5	Ensemble #2 (Bagged DT)	Bagging (base model: Decision Tree)
6	LogReg Backward	penalty=l2, C=1.0
7	KNN	n_neighbors=4
8	SVM	C=10, kernel=rbf
9	LogReg Forward	penalty=l2, C=1.0
10	Naive Bayes	Default parameters (GaussianNB)
11	Decision Tree	max_depth=10, min_samples_split=10

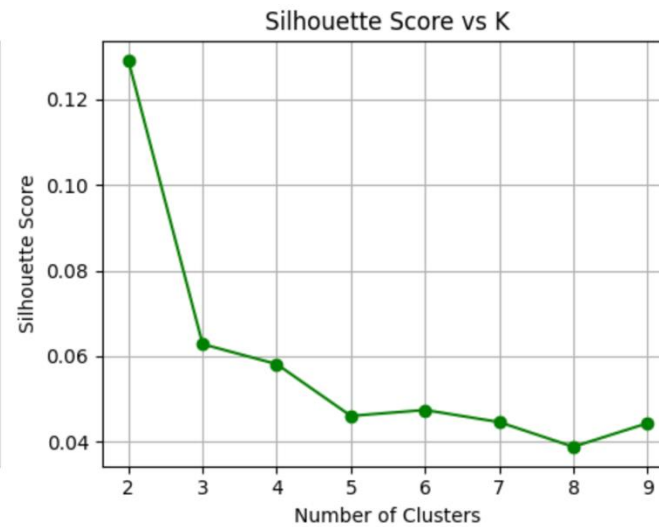
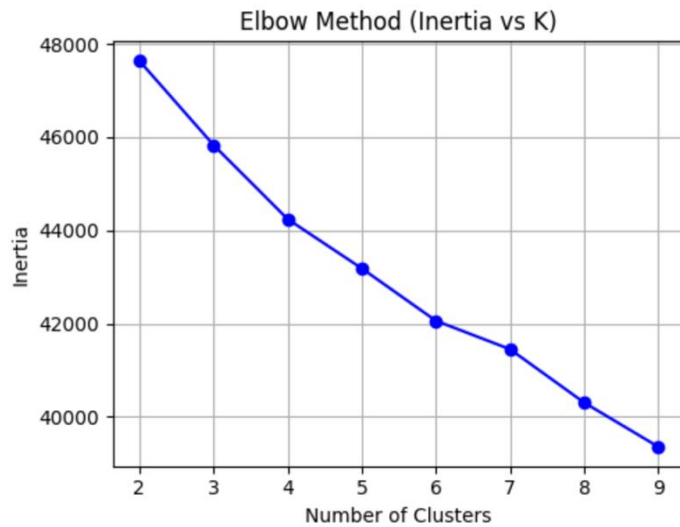
Classification Models Comparison



Models and Their AUC, Recall and Accuracy

Model	Accuracy	Recall	AUC
Gradient Boost	0.900	0.919	0.970
Extreme Gradient Boost	0.900	0.902	0.970
SVM	0.896	0.903	0.959
Random Forest	0.889	0.883	0.961
Ensemble #1 (SVM+DT+NN)	0.883	0.895	0.954
Ensemble #2 (Bagged DT)	0.862	0.866	0.934
Neural Network	0.858	0.883	0.941
Logistic Regression (Backward)	0.806	0.802	0.882
KNN	0.826	0.923	0.898
Decision Tree	0.802	0.862	0.815
Naïve Bayes	0.678	0.802	0.788

Clustering Models Interpretation:

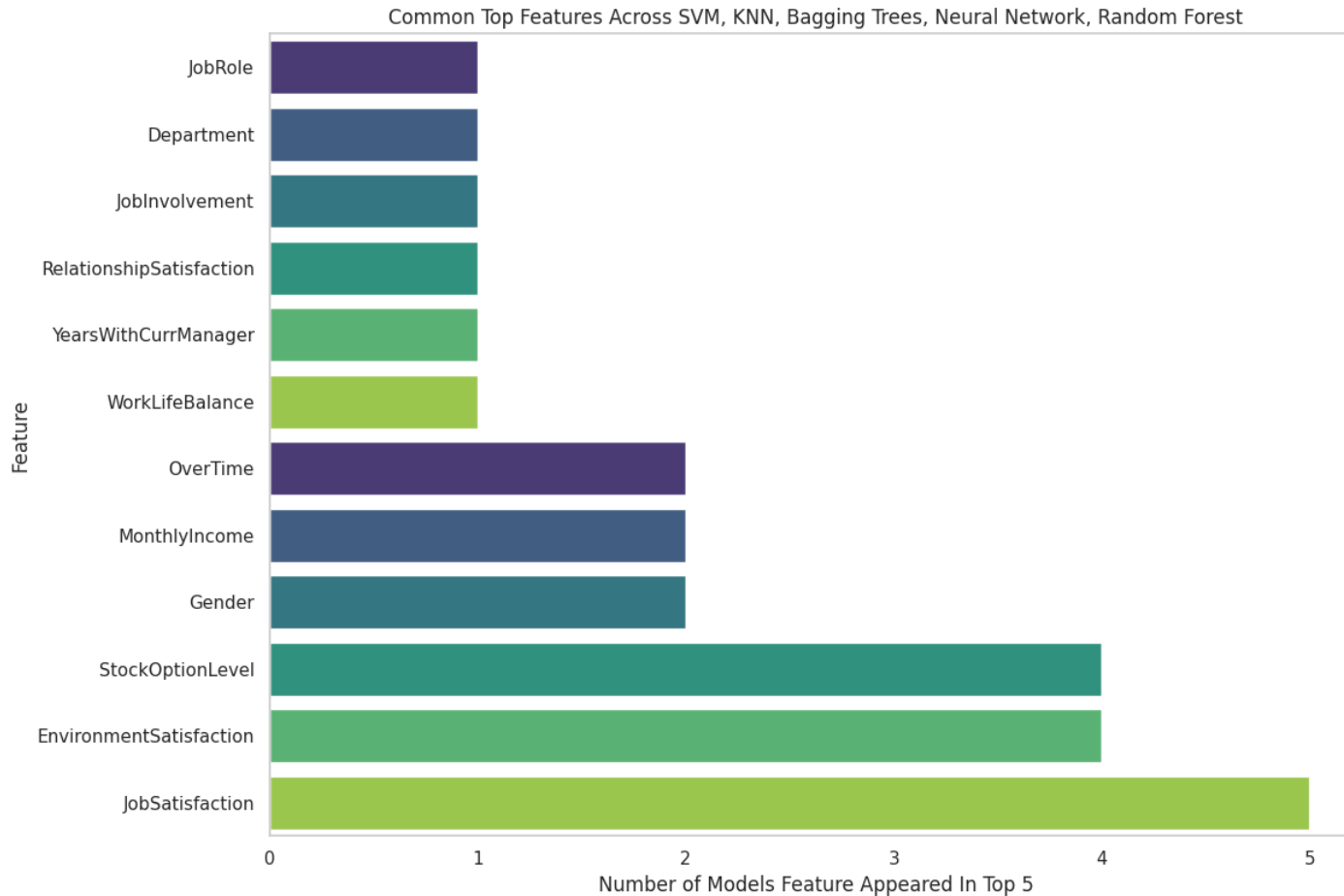


Aspect	KMeans	Agglomerative Clustering
Silhouette Score	0.1023	0.0726
Cluster Separation (PCA)	Better but still overlaps	Weaker, less clear
Interpretability	Good	Good
Best k identified?	Tentatively, k=2 or 3	Same

Interpretation of Clusters

Aspect	Interpretation with k=2
Natural Grouping	The dataset naturally divides into two distinct clusters.
Profile of Each Cluster	Cluster 0 and Cluster 1 differ significantly across key variables (e.g., age, travel distance, salary).
Segmentation Insight	These two groups can be used to design customized HR policies, marketing campaigns, or resource allocation strategies.
Validation	Silhouette Score and PCA 2D scatterplots help validate that k=2 provides reasonable separation between groups.
Comparison	Both KMeans and Hierarchical clustering support the 2-cluster structure, adding strength to the findings.

Top features and top model in predicting attrition



Top Performing Model:

Gradient Boost

- Accuracy: 90.0%
- Recall: 91.9%
- AUC: 0.970

Conclusion:

Attrition Risk Strategy

- Our predictive model identified job satisfaction, compensation, and manager relationships as the most critical drivers of attrition risk.
- Targeted action on these factors can significantly strengthen employee retention, reduce turnover costs, and enhance organizational performance.

